AI-Powered Job Recommendation System with Database Integration

INTERNSHIP PROJECT REPORT

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ABSTRACT

The AI Powered Job Recommendation System is designed to provide personalized job suggestions based on a user's skills, experience, and resume data. In today's competitive job market, traditional job portals often fail to deliver relevant results, as they do not consider individual user profiles effectively. This system addresses the problem by using machine learning techniques such as TF-IDF (Term Frequency–Inverse Document Frequency) and cosine similarity to analyze and match user input with job descriptions. Additionally, Natural Language Processing (NLP) is used to extract skills from uploaded resumes, enabling more accurate and automated profile building.

The system is implemented using Python and Flask, with SQLite as the backend database for storing user data, job information, and recommendation history. It also includes features such as secure user authentication, email-based password reset, skill gap analysis, and history tracking. By identifying both matched and missing skills, the system not only recommends suitable jobs but also helps users understand how to improve their profiles. Overall, the project demonstrates an effective application of machine learning and NLP in building an intelligent and user-friendly job recommendation platform.

AI-Powered Job Recommendation System with Database Integration

1. Problem Statement

In today's job market, finding suitable job opportunities that match a candidate's skills and experience is a major challenge. Most existing job portals provide general job listings without considering individual user profiles, leading to irrelevant recommendations and inefficient job searches. Users often struggle to identify which jobs are best suited for them and what additional skills are required to improve their chances of selection.

Therefore, there is a need for an intelligent system that can analyze user skills, experience, and resume data to provide personalized job recommendations. The system should be able to match user profiles with job requirements, identify skill gaps, and guide users toward better career opportunities using machine learning and natural language processing techniques.

2. Objectives

The main objectives of the AI Powered Job Recommendation System are:

- To develop a system that provides personalized job recommendations based on user skills and experience.
- To analyze user input and match it with job data using machine learning techniques.
- To extract skills automatically from resumes using Natural Language Processing (NLP).
- To calculate similarity between user profiles and job descriptions using TF-IDF and cosine similarity.
- To identify matched and missing skills for each job (skill gap analysis).
- To store user data, sessions, and recommendations using a database.
- To implement secure user authentication with login, signup, and password reset features.
- To maintain a history of recommendations for each user.
- To provide a user-friendly and interactive web interface.

3. Dataset Explanation

The dataset used in this project contains job-related information required for generating recommendations. It consists of multiple job records, where each record represents a specific job role along with its required skills and description.

The dataset is structured to support machine learning-based matching by providing relevant textual data for comparison with user input. Each job entry contains important attributes that help in identifying suitable matches for a candidate's profile.

Dataset Features

1. Job Title

Represents the name or designation of the job role, such as Data Scientist, Machine Learning Engineer, Data Analyst, etc.

2. Skills Required

This field contains the list of skills required for the job. These skills are used as the primary factor for matching with user skills.
Examples:

- Python
- Machine Learning
- SQL
- Deep Learning

3. Description

This field provides additional information about the job role, including responsibilities and requirements. It helps improve recommendation accuracy by giving more context to the model.

4. Methodology

The methodology describes how the system processes input data and generates job recommendations using machine learning and NLP techniques.

1. Data Collection

The job dataset containing job titles, required skills, and descriptions is collected and stored in the database. This data forms the base for generating recommendations.

2. Data Preprocessing

The dataset and user input are preprocessed to ensure consistency:

- converting text to lowercase
- removing special characters
- normalizing skill names
- handling missing values

3. Skill Extraction (NLP)

When a user uploads a resume:

- text is extracted from PDF or text file
- spaCy NLP model is used to identify skills
- PhraseMatcher detects predefined skill patterns
- RapidFuzz is used for approximate matching

4. Feature Extraction (TF-IDF)

- Text data (skills + description) is converted into numerical form
- TF-IDF assigns importance to each word
- Generates vector representation of text

5. Similarity Calculation

- Cosine similarity is used to compare user profile with job data
- Similarity score ranges from 0 to 1
- Higher score indicates better match

6. Recommendation Generation

- Jobs are ranked based on similarity score
- Top 5 jobs are selected
- Duplicate jobs are removed

7. Skill Gap Analysis

- Matched skills are identified
- Missing skills are calculated
- Helps users understand required improvements

8. Result Display

The system displays:

- job title
- match percentage
- matched skills
- missing skills
- recommendation status

9. Database Storage

- user input is stored in users table
- recommendations are stored in recommendations table
- history is maintained for each user

5. Algorithm Used

The system uses TF-IDF (Term Frequency–Inverse Document Frequency) and cosine similarity for generating job recommendations.

- TF-IDF converts textual data into numerical vectors by assigning importance to words.
- Cosine similarity measures the similarity between user profile and job descriptions.
- Higher similarity score indicates a better match between user skills and job requirements.

6. System Workflow

The Smart Food Waste Prediction System follows a structured workflow to generate predictions.

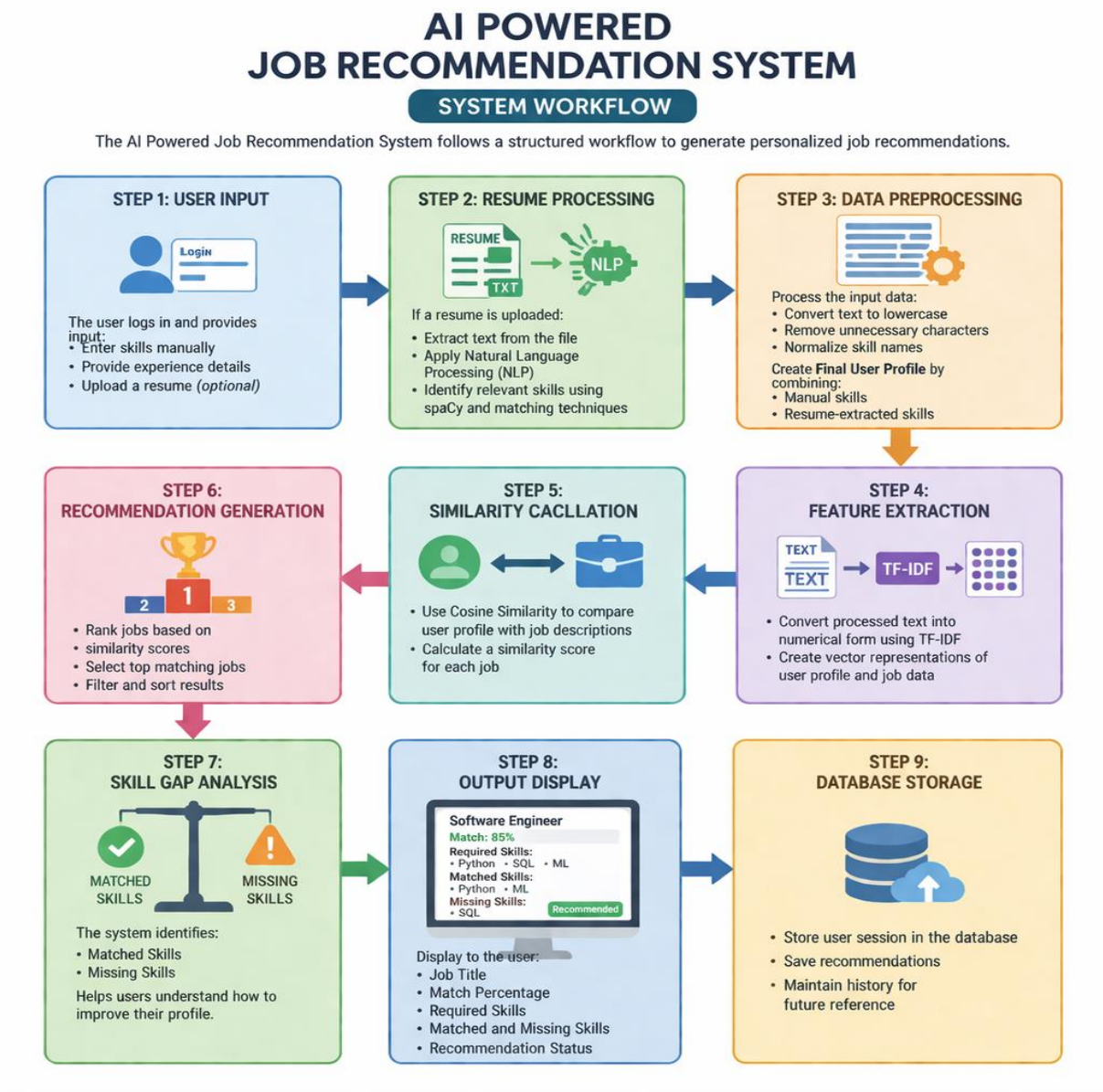

Figure 6.1: System Workflow of AI-Powered Job Recommendation

The diagram shows the workflow of the system, starting from user input (skills or resume), followed by NLP-based skill extraction, TF-IDF vectorization, and cosine similarity calculation. The system then generates job recommendations with match scores and displays matched and missing skills, while storing results in the database.

7. Results

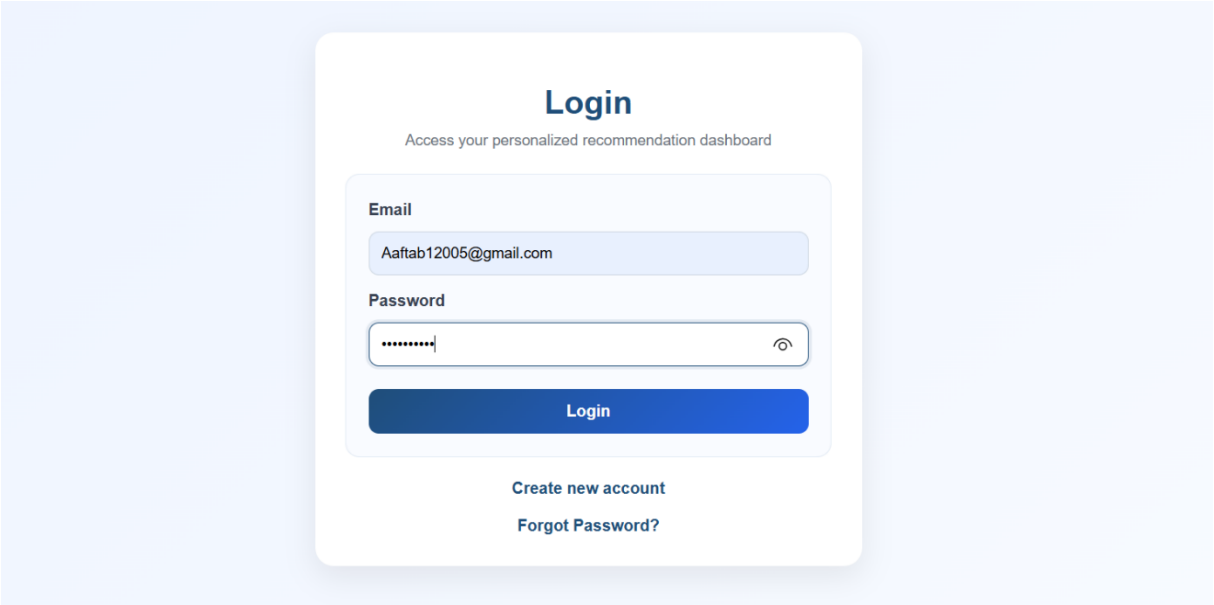

Figure 7.1 : Login page

This figure shows the login interface where users enter their email and password to access the system. It provides secure authentication and allows users to access their personalized dashboard.

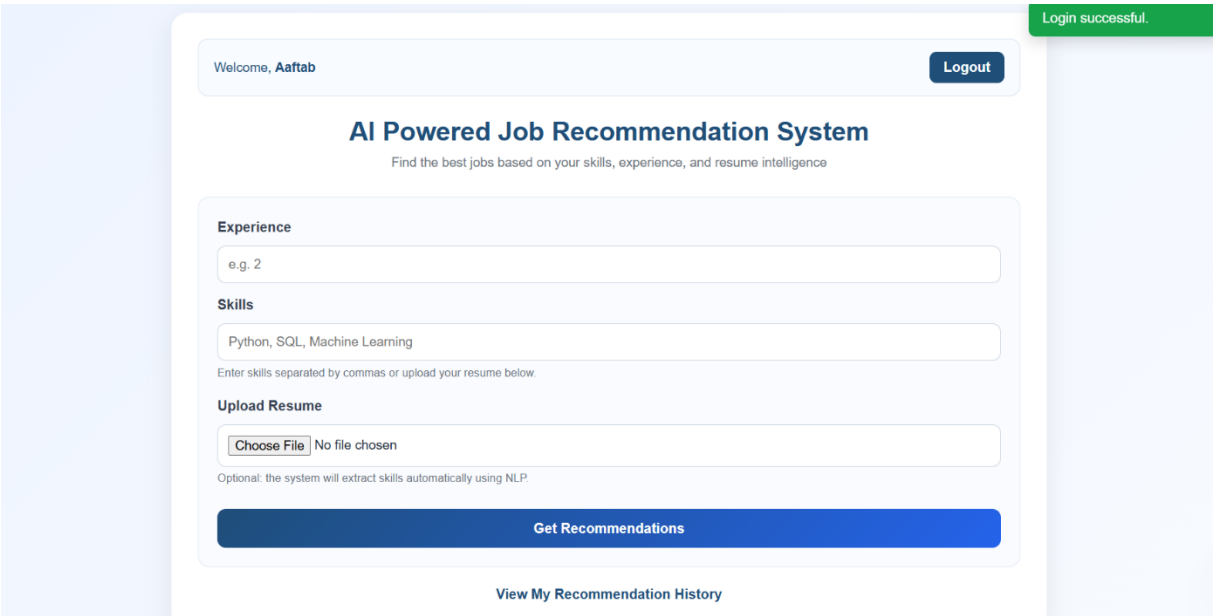

Figure 7.2 : User Interface

This figure displays the main dashboard where users input their experience, skills, and optionally upload a resume. The interface is designed to collect user data for generating job recommendations.

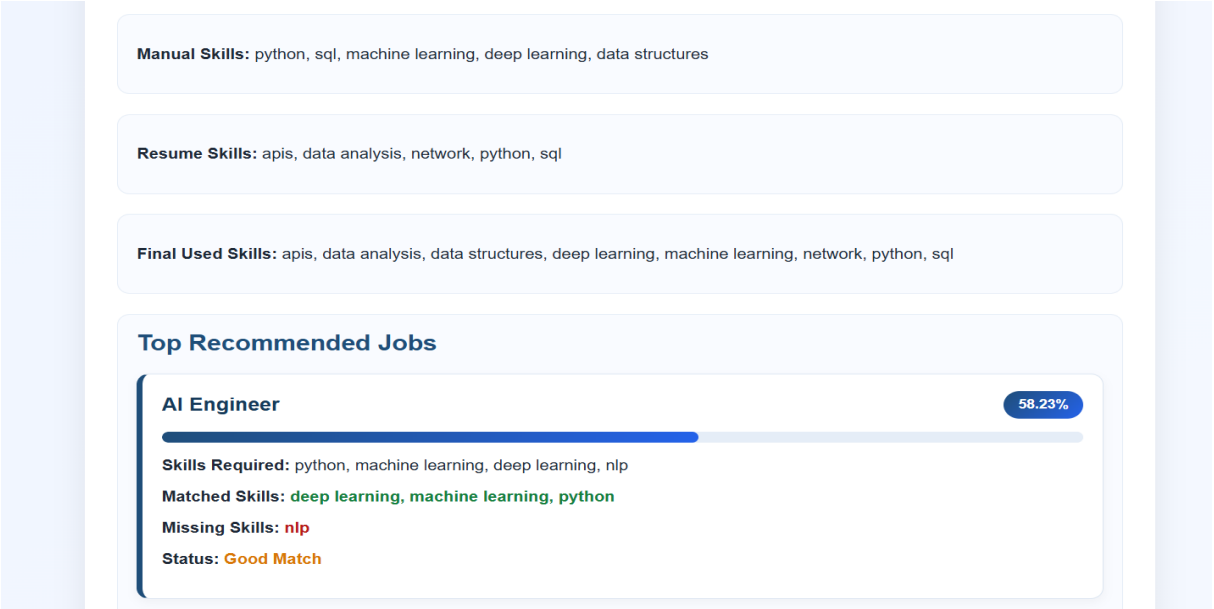

Figure 7.3 : Recommendation Output

This figure shows the output of the system, including extracted skills, final processed skills, and top recommended jobs. It also displays match percentage, required skills, matched skills, missing skills, and recommendation status.

8. Conclusion

The AI Powered Job Recommendation System demonstrates an effective use of machine learning and Natural Language Processing to provide personalized job suggestions. By analyzing user skills, experience, and resume data, the system is able to match candidates with suitable job roles using TF-IDF and cosine similarity techniques.

The system not only recommends relevant jobs but also performs skill gap analysis by identifying matched and missing skills, helping users understand how to improve their profiles. With features such as user authentication, recommendation history, and resume-based skill extraction, the project provides a complete and user-friendly solution. Overall, this system highlights the practical application of AI in career guidance and can be further enhanced with real-time job data and advanced models in the future.